Income Variables and Sources in the Canadian Community Health Survey Master File

1. Introduction

Income variables are disseminated on the Canadian Community Health Survey (CCHS) because of their potential association with health variables. The income variable most commonly used in analyses is total household income (variable INC_021). However, because of the sensitive nature of reporting income, in the past, the response rate to the total household income question was around 65% - 70% among CCHS respondents. In addition to the low response rate, a study conducted in 2009 suggested that income non-respondents had different health characteristics than the respondents. Furthermore, a study in 2015 found that respondent reported total household incomes may underestimate income (when compared to income tax records) and be less reliable for lower income households. In order to improve the quality of the income variables, while reducing the burden on respondents and saving on collection costs, linking to the tax data of respondents was introduced starting with the 2016 cycle of the CCHS[1].

Starting in 2016, the CCHS incomes variables represent a combination of tax data (~70%), collected respondent data (~15%) and imputed income amounts (~15%). The source used to populate the total income amounts for each respondent is identified by INCFIMPH for the household income (INC_021) and by INCFIMPP for the personal income (INC_041). Tables 1 and 2 below provide more information on the values of these two variables as well as the concordance to the previous flag, INCFIMP4.

Table 1. Values for INCFIMPH along with the correspondence to the 2015 flag.

INCFIMPH	2016 Code	2015 correspondence from INCFIMP4
Linked 2015 Tax Records	1	NA
Linked 2014 Tax Records	2	NA
Reported by Respondent	3	1
Imputed based on reported range	4	2 & 4
Imputed without income range	5	3

Table 2. Values for INCFIMPP

INCFIMPP	2016 Code
Linked 2015 Tax Records	1
Linked 2014 Tax Records	2
Reported by Respondent	3
Imputed based on reported range	4
Imputed without income range	5
Not Applicable	6

[1] Administrative tax records are used only on the master file. For more information on the share income sources, please see, "Income variables and sources in the Canadian Community Health Survey Share File".

For users interested in univariate analyses focusing on household income, it is recommended to rely on other income data sources than the CCHS. Those interested in analysing household income as it relates to health should use derived income variables such as income quintiles rather than the direct income value. The source used to populate the income amounts have slight differences and varying degrees of reliability. The imputation process performs well at predicting income quintiles but there is more variable in estimating the exact income value.

This document gives a general overview of the income sources as well as the imputation process.

2. Tax Data

Respondents not refusing the linkage to tax data and for whom tax records could be linked, have their sources of income (INC_005A-O, INC_Q025A-O), their main source of income (INC_015, INC_035) and the total income amounts (INC_021, INC_041) determined from their tax records. Tax records were used both at the personal and household level. While only ~70% of respondents' CCHS data were linked to tax records, those linked were of a very high quality. The estimated false positive rate was 0.3%, meaning that about three out of every thousand respondents were incorrectly matched to tax information.

Tax records stem from one of two sources. Tax records from 2015 were used where possible. This corresponds to the reference year used by respondents in the cases of reported incomes and sources. Where a link could not be established for 2015, 2014 tax data was used. In addition to the change in year, the 2014 files have had more processing and have slightly different concepts as a result. The 2014 file groups people by the concept of "family" and not by "household". Income sources and amounts in some cases may not be based on the same set of persons for the 2014 files as with the 2015 files.

Users should be aware of differences arising between data reported by respondents and income data from tax records. In the CCHS, the size of the household is determined based on the number of occupants rostered by the interviewer at the time of the interview. The reference period for respondent reported incomes is 2015 and either 2015 or 2014 for the linked tax records. In some cases, the household composition or size may have changed between the reference period for incomes and the time of interview. The total household incomes are based on the linked tax filers for the tax reference year, not the CCHS determined household size in 2016.

Internal studies comparing the CCHS respondent income data to linked tax records have highlighted other key differences. On average, respondents slightly underestimate the total income, as compared to the incomes filed on their tax return. Therefore, total household and personal incomes will be on average higher than what would be expected if the respondent answered the questions. Furthermore, the tax records show a more comprehensive picture of income sources. Sources with small income amounts such as those from Dividends and Interest (INC_005C, INC_025C) are found in tax records but are much less commonly reported by respondents. Income from Social Assistance/Welfare (INC_005J, INC_025J) is also more prevalent in tax records than it is reported.

3. Respondent Data

Income questions were asked of respondents where selected households were not deemed linkable to tax data at the time of sampling or in cases where respondents refused to allow the data linkage. The respondent data is used where linkage to tax data was not feasible. As noted above, discrepancies

between collected respondents data and their tax records have been identified in studies. Respondents tend to miss reporting some sources of income such as Dividends and Interest (INC_005C, INC_025C) and Social Assistance/Welfare (INC_005J, INC_025J). Reported incomes are on average lower than the income found in linked tax records. Reported total incomes are often also rounded. Where possible, tax records were used instead of respondents' data.

4. Income imputation

In about 15% of cases, incomes were imputed for the total household income and the total personal income (where applicable). Income imputation was performed for total income amounts where there was no collected data and links to tax data were either not found or linking was refused by respondents. Only total income amounts were imputed, not sources of income. Where the total income has been imputed for INC_021 or INC_041, the sources of income questions (INC_005A-O, INC_Q025A-O), and main source of income (INC_015, INC_035) questions are populated if the respondent provided responses and are not stated otherwise.

4.1 Overview of the Imputation Process

A nearest neighbour imputation method is used to impute income. The modeled household and personal income defined in the following section is used as a distance measure to determine which pair of donor-recipient is the "nearest" within imputation classes.

4.1.1 Modeled Income

An important auxiliary variable used in the imputation process is **modeled income.** It is computed for all CCHS respondents (whether they responded to the income questions or not) using a three step process. Details are given below.

<u>Step 1</u>
Using a regression model, personal income of *all* members in a responding household is derived. Personal total income of the selected respondent and of any other household member are derived separately. Personal total income of respondents is first derived using variables including age group, education level, sources of income, province of residence, marital status, household size, whether the dwelling is owned or rented, immigrant status, and health status such as whether the person has cancer.

Personal income of all other members is derived using the same variables as above but excluding immigrant status and the personal health information.

For total personal income, step 2 below is skipped and the value imputed for just the CCHS respondent is used to determine the nearest neighbor in the imputation process

<u>Step 2</u>
A preliminary household income value is derived by summing all personal incomes from step 1. The resulting totals do not accurately reflect the income information collected by the CCHS so these values require further adjustments.

Step 3

The median household income from T1 tax data is derived for each postal code by household size group. Using a regression model with the median income from tax, health variables and the preliminary income from step 1 for personal income and from step 2 for household income, a final modeled income is derived. This final modeled income does not appear on the final data file, but is rather used as an auxiliary variable in the nearest neighbour imputation process for total income.

4.1.2 Choosing a Donor

Imputation classes are formed differently depending on whether the income being imputed is a total household income or total personal income as well as whether respondents provided an income range or not. For total household income and those providing an income range, imputation classes are based on the income range and household size. For people who do not provide an income range, imputation classes for total household income are formed based on health region and household size. For total personal income, imputation classes are based on age and sex in lieu of the household size.

5. Data File

Users can use the income flags of INCFIMPH and INCFIMPP to identify the source of the income variables and their quality. Derived income variables such as INCDHH, INCDADR, INCDRCA, etc., will be based on the imputed household income value where an imputation was performed.

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